

Skills Summary

Squared or Cubed? — Units Guide

Use this reference while completing your worksheet or when studying.

Skill 1 — Square Units Measure Flat Space

Rule: length $\rightarrow$ cm • flat surface (area) $\rightarrow$ cm² • filled space (volume) $\rightarrow$ cm³.

Area of a rectangle: $A = \ell \times w$ — it counts how many unit *squares* cover the surface, so its unit is always *squared*.

Example: Find the area of a 9 cm by 5 cm card.

Step 1. A card's face is a flat surface, so we count unit squares — multiply the two side lengths and report square centimeters.

Step 2. $A = 9 \times 5 = 45$ unit squares.

$$\Rightarrow A = 45 \text{ cm}^2$$

Try it: Find the area of an 8 cm by 3 cm bookmark.

check: 24 cm²

Skill 2 — Cubic Units Fill Solid Space

Rule: volume of a box: $V = \ell \times w \times h$ — it counts how many unit *cubes* fill the space, so its unit is always *cubed*. Three lengths multiplied $\rightarrow$ the unit is used three times.

Example: Find the volume of a $4 \times 3 \times 2$ cm box.

Step 1. The box holds stacked layers of unit cubes, so multiply all three dimensions and report cubic centimeters.

Step 2. Base layer: $4 \times 3 = 12$ cubes; 2 layers make 24 cubes.

$$\Rightarrow V = 24 \text{ cm}^3$$

Try it: Find the volume of a $5 \times 2 \times 2$ cm box.

check: 20 cm³

Skill 3 — Converting Squared Units

Rule: squared units use the conversion factor *twice*. Each meter is 100 cm, so

$$1 \text{ m}^2 = 100 \times 100 \text{ cm}^2.$$

(Cubed units would use it three times.)

Example: Convert 3 m² to square centimeters.

Step 1. A square meter is a 100 cm by 100 cm square, so the factor 100 appears twice — once for each dimension.

Step 2. $3 \times 100 \times 100 = 3 \times 10000 = 30000$.

$$\Rightarrow 3 \text{ m}^2 = 30000 \text{ cm}^2$$

Try it: Convert 5 m^2 to square centimeters.

Check: 50000 cm^2

(!) Watch out: a correct *number* with the wrong *unit* is a wrong answer. Before writing the unit, ask: am I measuring a length, a flat surface, or a filled space?